Stochastic Loewner Evolution

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1. Motivation: criticality of 2D systems in and out of equilibrium

It is no news that various models of statistical mechanics exhibit phase transitions as a control parameter varies (usually the temperature T or a probability p governing the microscopic dynamics) and critical points, in the neighbourhood of which various macroscopic quantities are characterised by power-law dependences $X \sim |t|^{\beta}$. A peculiarity of many models in (embedding) dimension $D = 2$ is that at the critical point they can be characterised by self-avoiding curves with peculiar features, for example endowed with fractal dimension > 1.
Two important examples in equilibrium physics are the Ising model and isotropic lattice percolation, in which it is possible to think of every configuration as an assignment of $\pm$ to every vertex of the lattice. At the critical point of these models one observes the formation of large clusters of $+$ and of $-$ (whose characteristic size scales with a power law that defines a critical exponent). The separation curve between two clusters is self-avoiding by construction, and generally has fractal behaviour. Another important example coming from polymer physics is the *self-avoiding walk*, i.e. a random walk in which the increments are not independent but a single walk is required never to intersect itself. This model too generates self-avoiding curves, this time by definition. Regarding the analysis of these curves in equilibrium models at the critical point, three crucial observations can be made:

1. Even if the model has a discrete lattice, it is always possible, through a coarse graining process, to make it continuous, so as to really have continuous curves rather than a union of segments (edges of the lattice);

2. The models described are all probabilistic models, in which each configuration is endowed with a certain probability (for example in the Ising case it will be the typical Boltzmann probability). This also induces a probability on the curves, seen as particular events in the configuration space;

3. The fractal properties of the curves come from the fact that many equilibrium models at the critical point exhibit conformal invariance, i.e. they remain invariant under rotations and local dilations. This gives the curves self-similarity properties, a factor that induces non-trivial fractal dimensions.

A question that may be interesting to ask is whether it is possible to characterise the probability space of these curves knowing the starting model and assuming that at the critical point there is conformal invariance. An important result, and the main subject of this text, comes from the stochastic Loewner equation (abbreviated SLE), which manages to map probability problems for curves into a stochastic process in 1d, which is shown to be necessarily a Brownian motion $\sim \sqrt{k}B_t$ parametrised by a real variable k. The interesting fact about this construction is that all the examples mentioned above satisfy this equivalence, and among them only the specific value of the parameter k changes. The aim of this text is to provide an outline of proof (in physics style) of the construction of SLEs, showing some examples of application. Finally, we will discuss the use of SLEs for anisotropic non-equilibrium systems in which conformal invariance is not satisfied, trying to understand from numerical simulations which adaptations must be faced in these cases.

2. Loewner equation and conformal mapping

2.1 Mapping and uniqueness

The first step in the construction of SLEs is to be able to map non-intersecting curves in a 2D space into a 1D line. Being in 2 dimensions, it is convenient to think of the plane as the complex plane $\mathbb{C}$ and to use all the properties coming from complex analysis. To study the behaviour of the curves, we consider in particular the upper half-plane $\mathbb{H}$ and restrict ourselves to curves (denoted from now on by K_t) growing from the origin. Since they are self-avoiding, the domain $\mathbb{H}/K_t$ is simply connected, like the original $\mathbb{H}$. At this point we can make use of a mapping theorem for simply connected domains due to Riemann.

Theorem 2.1: Riemann mapping

Given two simply connected sets different from $\mathbb{C}$, D, D', there exists a biholomorphic function $f : D \to D'$.

In the case under analysis, the theorem tells us that there exists a holomorphic function $g_t : \mathbb{H}/K_t \to \mathbb{H}$ with holomorphic inverse $f_t : \mathbb{H} \to \mathbb{H}/K_t$. To be precise, for every "instant of growth" (time t) we have these two functions, which map the tip of the curve to a point of the real axis.
Example: If $K_t = \{0 \leq Im(z) \leq L\}$, the map $g_t(z) = (z^2 + l^2)^{1/2}$ maps $(0, L) \to (0, 0)$, the right side of the stick into the real interval $(0, L)$ and real points into other real points.
Regarding the uniqueness of such transformations, we observe that it is possible to define a family of self-holomorphisms $\mathbb{H} \to \mathbb{H}$ as $h(z) = a + b/(z - c)$, where a, b, c are real constants. To define canonically a unique transformation it is enough to fix the behaviour of g_t at infinity, requiring that it coincide with the identity:

$$g_t(z) = z + O(1/z). \tag{2.1}$$

2.2 The case of curves in $\mathbb{H}$

Let us now see how to specialise the previous equation, looking for an equation that determines, as a parameter t varies, the relation between the maps g_t and the position of the image of the tip on the real axis a_t. From Cauchy's integral representation theorem we can write the inverse of g_t, f_t:

$$f_t(w) = w - \int_{\mathbb{R}} \frac{\rho_t(x')\mathrm{d}x'}{w - x'}. \tag{2.2}$$

Where w is isolated so as to make the boundary term at infinity vanish. By holomorphy of f_t, $\bar{f}_t(w) = f_t(\bar{w}) \to \rho(x') \in \mathbb{R}$, and since the domain of f_t is $\mathbb{H}$ one sees that $\rho_t(x') \geq 0$ (in particular $\rho_t(x')\mathrm{d}x'$ is a positive measure). We can now define explicitly a time, called *Loewner time*[1]:

$$t = \frac{1}{2}C_t = \frac{1}{2}\int_{\mathbb{R}} \rho_t(x')\mathrm{d}x'. \tag{2.3}$$

[1]Another choice that might seem sensible consists in taking the length of the curve as time. This choice, however, may not prove convenient when the curve is fractal (a property which, on the other hand, one is trying to obtain).

With this choice one sees that asymptotically the time is additive:

$$f_t(w) \sim w - \frac{C_t}{w}; \tag{2.4}$$

$$g_t(z) \sim z + \frac{C_t}{w}; \tag{2.5}$$

$$g_{t_2}(g_{t_1}(z)) \sim z + \frac{2t_1}{z} + \frac{2t_2}{z}. \tag{2.6}$$

Considering the evolution of the time parameter from t to $t' = t + \mathrm{d}t$, we have a chain of inclusions for the sets $K_t \subset K_{t'}$ (a consequence of the fact that we assume a growing curve). Now composing the transformations $g_{t'} = \delta g_t \circ g_t$ one obtains the relation:

$$g_t(z) = g_{t+\mathrm{d}t}(z) - \int_{\mathbb{R}} \frac{\delta\rho_t(x')\mathrm{d}x'}{g_{t+\mathrm{d}t}(z) - x'}. \tag{2.7}$$

Which, turning it around and assuming that the curve grows infinitesimally in time ($\delta\rho_t \sim \delta t \nu_t$):

$$\frac{d}{dt}g_t(z) = \int_{\mathbb{R}} \frac{\nu_t(x')\mathrm{d}x'}{g_t(z) - x'}. \tag{2.8}$$

Note that with the definition of Loewner time we would have $\int \nu_t(x')\mathrm{d}x' = 2$. We observe that we still have freedom of choice on the measure ν_t. Requiring that the curve always grows from its tip (and that it has no branches) one obtains $\nu_t(x) = 2\delta(x - a_t)$, where a_t is the image of the tip under g_t on the real axis. Substituting, one obtains the *Loewner equation*:

$$\boxed{\frac{d}{dt}g_t(z) = \frac{2}{g_t(z) - a_t}.} \tag{2.9}$$

The Loewner equation does exactly what we wanted: through a differential equation it links bijectively a continuous function on $\mathbb{R}$, a_t, to a curve in $\mathbb{H}$ through the function $g_t(z)$, and vice versa.

Example: Going back to the previous example of the stick, one sees that it is obtained by imposing $a_t = 0$. In particular one finds $g_t(z) = \sqrt{z^2 + 4t}$. Conversely, from the construction of the stick, one could observe that since it does not move horizontally one had to have $a_t = 0$ by symmetry.

Note that so far we have assumed only that the curve is continuous and has no intersections. In reality this condition is not necessary, and sometimes it is not satisfied. What really matters is to consider a simply connected domain $\mathbb{H}/K_t$, not that K_t be a simple curve (although in that case it is immediate that the complement is simply connected). For example it can happen that the curve touches the real line, giving rise to a bubble whose boundary is the curve and a segment of the axis. Looking at the Loewner equation one has $g_t(a_t) = a_t$, a fact that generates a divergence in δg_t. What happens physically is that the whole bubble must be swallowed into K_t, an action that causes a finite increase of mass in an infinitesimal time. Note that the self-intersection of the curves comes from the fact that the size of the lattice edges has been sent to 0 (in the case of a lattice of finite size I actually have a self-avoiding curve). For this reason, when it intersects itself it always does so tangentially.

3. Stochastic Loewner equation

3.1 introduction of stochasticity

Let us now see where the notion of stochasticity emerges in the problem. As we have already discussed, the goal of this construction is to determine properties of curves present in models of statistical mechanics. In these models a probability is defined (usually an equilibrium one), which induces a probability on the formation of the various non-intersecting curves. Once the probability space of these curves is considered, the function g_t transfers the notion of randomness onto the function a_t, which becomes a 1d stochastic process. Finally, note that talking about a stochastic phenomenon in 1d of real functions is, at least conceptually, much simpler than talking about the measure space of conformal curves in $\mathbb{H}$. That said, we have not yet characterised the precise stochastic process that defines a_t. To do so it is necessary to use the last missing ingredient...

3.2 Conformal invariance and Brownian motion

The time has come to use the conformal invariance hypothesis of the statistical model in question. Note that this assumption can be made only at the critical point and is strongly model-dependent. The reasoning that leads to the determination of a_t is the following. Consider a curve which, starting from the origin, has grown up to a time t, and let us ask what the probability space looks like that, given the past up to t, it has a certain well-defined shape when observed at time s, in the domain $\mathbb{H}$. This will be equal to the probability that the curve develops a certain shape from t to s, in the domain $\mathbb{H}/K_t$[2]. By conformal invariance, if we apply an angle-preserving transformation these curves will be mapped into new curves, which will keep the original probability. The conformal map we choose to apply is the holomorphic function g_t, which sends $\mathbb{H}/K_t \to \mathbb{H}$, the tip to a_t (which is a point of the real line), the curve up to time t onto the real axis, and the curve from t to s to a curve in $\mathbb{H}$. What do we learn from this?

- the probability space of the increments of the curve can be mapped into a space of curves that have no dependence on the past. If we take infinitesimal increments, this tells us that they are independent. In particular the process is Markovian in the stochastic sense;
- Since it is possible to translate $a_t \to 0$ without changing the space (translation is a conformal map), the increments are also stationary.

The independence and stationarity of the increments on the curve translate into independence and stationarity of the increments of the 1d process a_t. We have therefore found that the stochastic process a_t is a continuous process with stationary and independent increments. This forces it to lose a lot of generality; in particular it takes the form:

$$a_t = \sqrt{k}B_t + \alpha t. \tag{3.1}$$

Where B_t is the Brownian motion with variance t. Considering that the models are isotropic, it is possible to eliminate the Drift term ($\alpha = 0$), so that the result we were looking for remains:

$$\boxed{a_t = \sqrt{k}B_t.} \tag{3.2}$$

[2]This equivalence is called geometric Markovianity.

We observe that k is the only parameter able to allow a distinction between the different models through this mapping. For the models mentioned above we have:

- For the boundaries of Ising clusters $k = 3$;
- For the boundaries of percolation clusters $k = 6$;
- For the self-avoiding walk $k = \frac{8}{3}$.

SLE_k is the name of the family of conformally invariant curves from 0 to ∞, in $\mathbb{H}$, described by equation (2.9), where a_t is understood as a realisation of the Brownian motion. We note finally an interesting result: although the measure of these curves is mapped into a Markovian stochastic process, the process by which these curves are built is strongly non-Markovian, for example in the case of self-avoiding walks. Still regarding the simplicity property of the curves, one observes that for $k < 4$ one obtains simple curves, while for $k > 4$ the curves are self-intersecting.

4. Application: fractal dimension

Let us see a couple of examples of how it is possible to use SLEs to infer information about the curves. A first quantity that can be estimated exactly is the fractal dimension of a model characterised by the associated value of k. To do so one considers that in general the fractal dimension of a curve can be obtained from the probability that it intersects a circle of radius ϵ centred at r, through the estimate:

$$P(r, \epsilon) \sim \epsilon^{2-d_f} f(r). \tag{4.1}$$

Where d_f is the fractal dimension and $f(r)$ some function of r. By conformal invariance this probability is the same as that of the problem obtained by applying the holomorphic function $\hat{g}_{\delta t}(z) = g_{\delta t}(z) - a_{\delta t}$, considering the circle of radius modified to $|g'_{\delta t}(r)|\epsilon$. From the Loewner equation one obtains:

$$P(x, y, \epsilon) = \left\langle P\left(x + \frac{2x\,\delta t}{x^2+y^2} - \sqrt{\kappa}\,\delta B_t,\ y - \frac{2y\,\delta t}{x^2+y^2},\ \left(1 - \frac{2(x^2-y^2)\,\delta t}{(x^2+y^2)^2}\right)\epsilon\right)\right\rangle. \tag{4.2}$$

Where the average is taken over the Brownian motion defined by a_t. Expanding for the infinitesimal transformation one finds:

$$\left(\frac{2x}{x^2+y^2}\frac{\partial}{\partial x} - \frac{2y}{x^2+y^2}\frac{\partial}{\partial y} + \frac{\kappa}{2}\frac{\partial^2}{\partial x^2} - \frac{2(x^2-y^2)}{(x^2+y^2)^2}\epsilon\frac{\partial}{\partial \epsilon}\right) P = 0. \tag{4.3}$$

Which for $k \leq 8$ has solution:

$$P \sim \epsilon^{1-\kappa/8}\, y^{(\kappa-8)^2/8\kappa}\, (x^2+y^2)^{(\kappa-8)/2\kappa}. \tag{4.4}$$

So that $d_f = 1 + k/8$. For $k \geq 8$ one obtains $d_f = 2$. The general solution to the problem is therefore:

$$\boxed{d_f = min(1 + k/8, 2).} \tag{4.5}$$

5. Non-equilibrium case: what can we learn?

In the previous sections we have seen how it is possible to investigate properties of statistical systems at the critical point that exhibit conformal invariance. There are cases in which this assumption cannot be made even approximately. One of the best known is the directed percolation model, which can be understood both as a non-equilibrium process in 1D and as a statistical process in 2D. In this case the time direction and the space direction are substantially different from each other, and for this reason the 2D model has no rotation invariance, hence no conformal invariance. Nevertheless, although one can no longer assume that a_t follows the law of Brownian motion, the Loewner equation (2.9) continues to hold, as does the observation that a_t must be a continuous stochastic process. At this point two different analyses can be carried out experimentally:

1. One simulates the problem under study and maps the curves into the process a_t through g_t. At this point one infers the properties that the 1D process must have;

2. One models a_t with a precise stochastic process, different from Brownian motion, and sees which system of curves it generates through g_t^{-1}.

Applying the first strategy to the directed percolation model (as well as to other anisotropic systems) and computing the mean square displacement of the real process, one finds a relation of the type:

$$\boxed{\langle X^2 \rangle = bt^{\alpha}.} \tag{5.1}$$

Where in the case of directed percolation $b = 3.74 \pm 0.07$ and $\alpha = 0.676 \pm 0.001$. The stochastic process underlying directed percolation is therefore characterised by an anomalous diffusion, i.e. by $\alpha \neq 1$. Having found this, through the second approach it is possible to study known processes of anomalous diffusion and see whether they reproduce anisotropic models such as DP. One of the simplest processes is fractional Brownian motion, $B_H(t)$, where H is a number called the Hurst index, characterised by the following properties:

- The increments ΔB_H are stationary (but not independent!);
- the time correlation $\mathbb{E}[B_H(s)B_H(t)] = \frac{1}{2}(t^{2H} + s^{2H} - |t-s|^{2H})$
- for $H = 1/2$ one obtains standard Brownian motion;
- the Hausdorff dimension is 2-H (if it is embedded in 1d);
- $B_H(at) \sim |a|^H B_H(t)$.

Comparing this process with the one induced by DP, the same trends are observed both as regards the (anti)correlation and the production of anisotropic curves.
Finding a precise mapping between stochastic processes and curves in non-isotropic systems such as those produced by non-equilibrium dynamics is an activity that still requires time, although comparisons such as those carried out through numerical simulations suggest that there is hope for future progress.

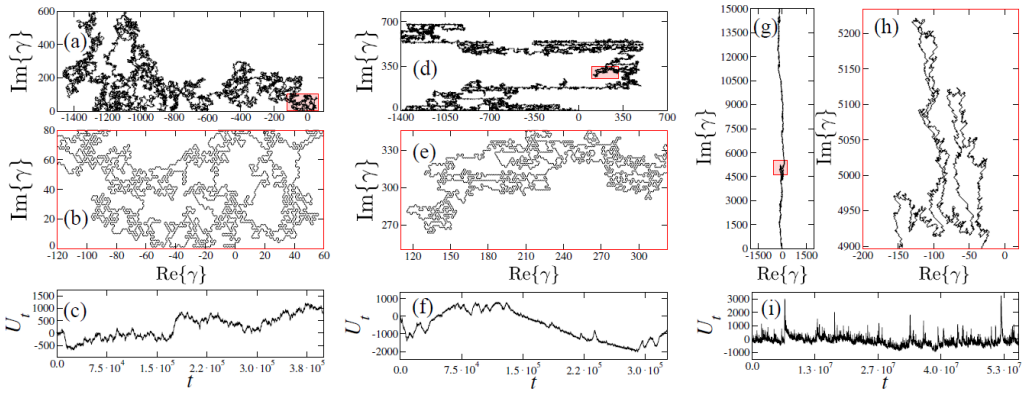
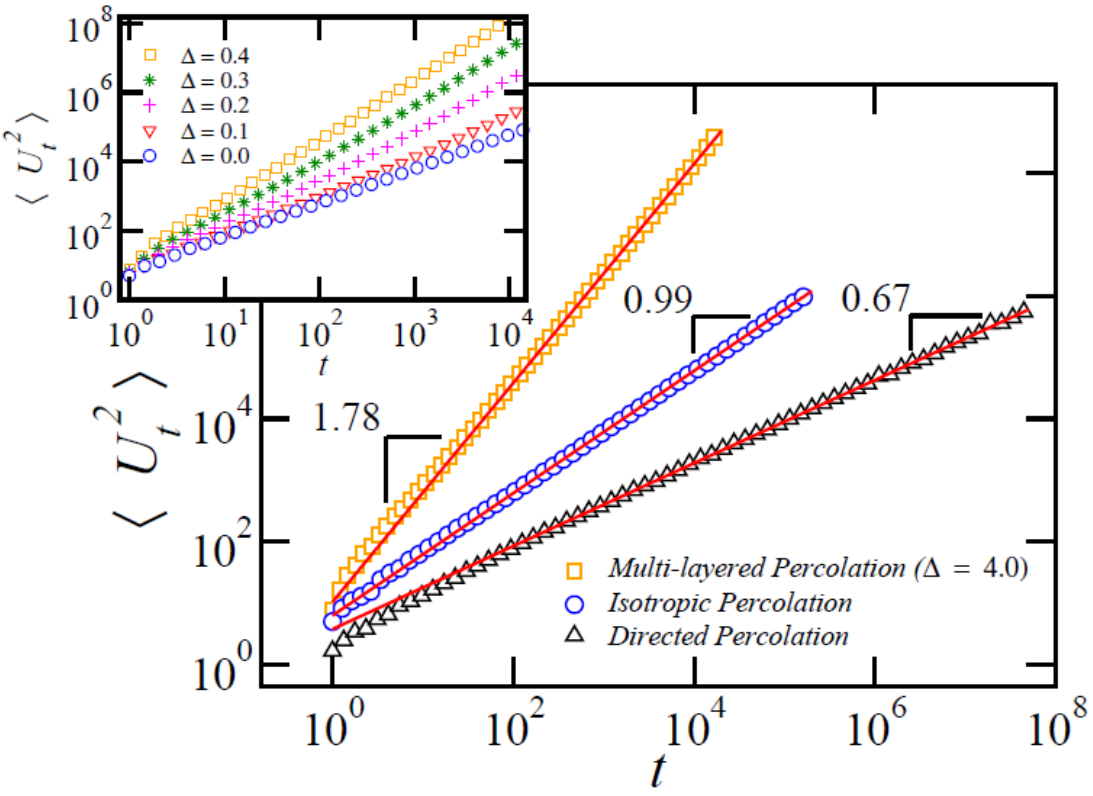


FIG. 2. Examples of cluster perimeters at the critical point of (a) isotropic percolation on a triangular lattice ($p_c = 0.5$), (d) multi-layered percolation also on a triangular lattice ($\Delta = 0.2$ and $p_c = 0.5$), and (g) directed percolation on a square lattice ($p_c \approx 0.644$). A detail of each curve in (a), (d) and (g) (pink square) can be seen in (b), (e) and (h), respectively. The driving functions obtained by applying the zipper algorithm [18] to the curves (a), (d) and (g) are shown in (c), (f) and (i), respectively.



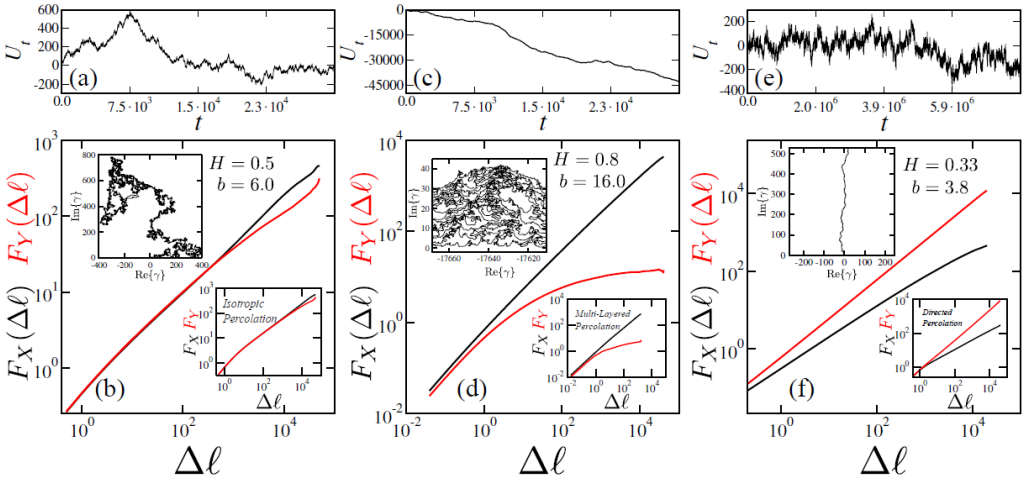


FIG. 4. Root mean squared estimations of the displacements in the X and Y-directions of SLE traces driven by long-range power-law correlated time series (fractional Brownian motion). In (a), (c) and (e) we show typical realizations of uncorrelated, correlated and anti-correlated driving functions, respectively. The simulation parameters (H, b and t_f) were chosen based on the results shown in Fig. 3 (see Table I for the numerical values). Good agreement is observed between the uncorrelated result (b) and isotropic percolation (inset on the bottom), as it is expected. The correlated trails (d) are also compatible with multi-layered percolation (inset on the bottom). In the anti-correlated case (f), the same kind of anisotropy present in the directed percolation is observed (inset on the bottom). These results support our hypothesis that long-term correlations in the driving functions, i.e., the presence of anomalous diffusion, are responsible for the anisotropic behavior of the traces. The insets on the top of (b), (d) and (f) show examples of the traces generated from the simulations with the corresponding driving function shown in (a), (c) and (e), respectively.

6. References

- Lectures on Stochastic Loewner Evolution and Other Growth Processes in Two Dimensions, by John Cardy, for the introduction to SLEs and applications;
- Article: Stochastic Loewner Evolution Relates Anomalous Diffusion and Anisotropic Percolation, for the approach to the DP model.